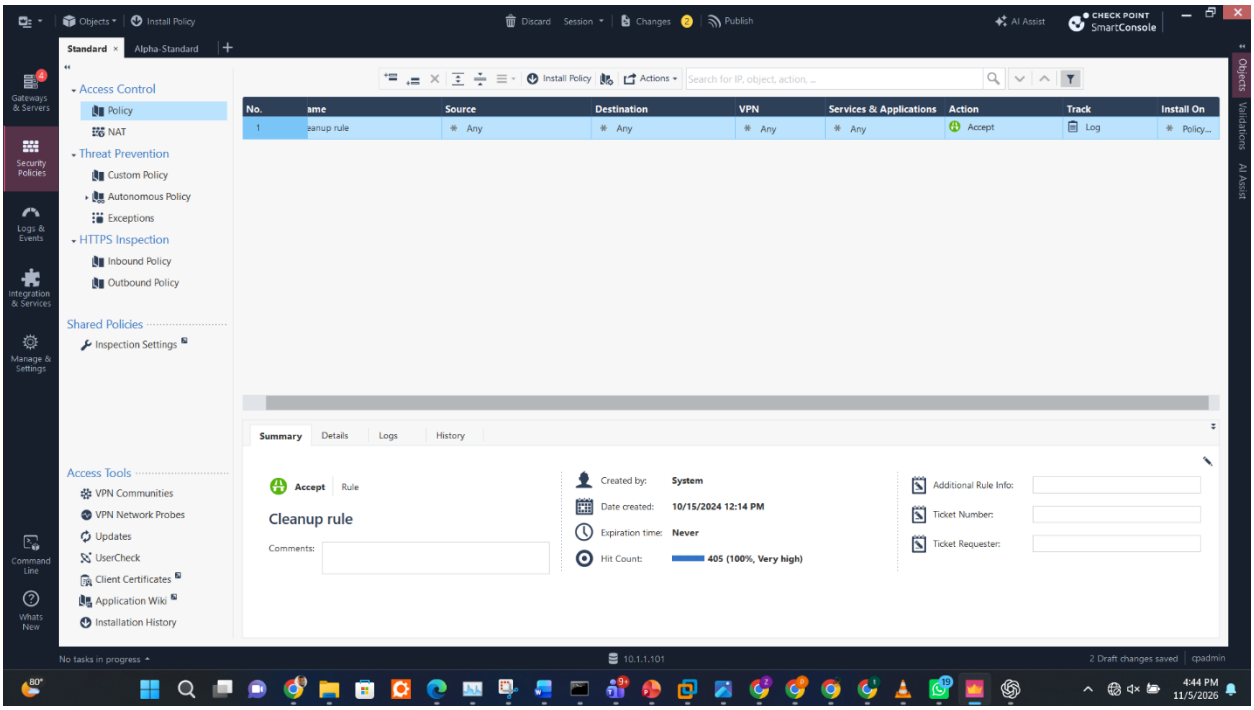
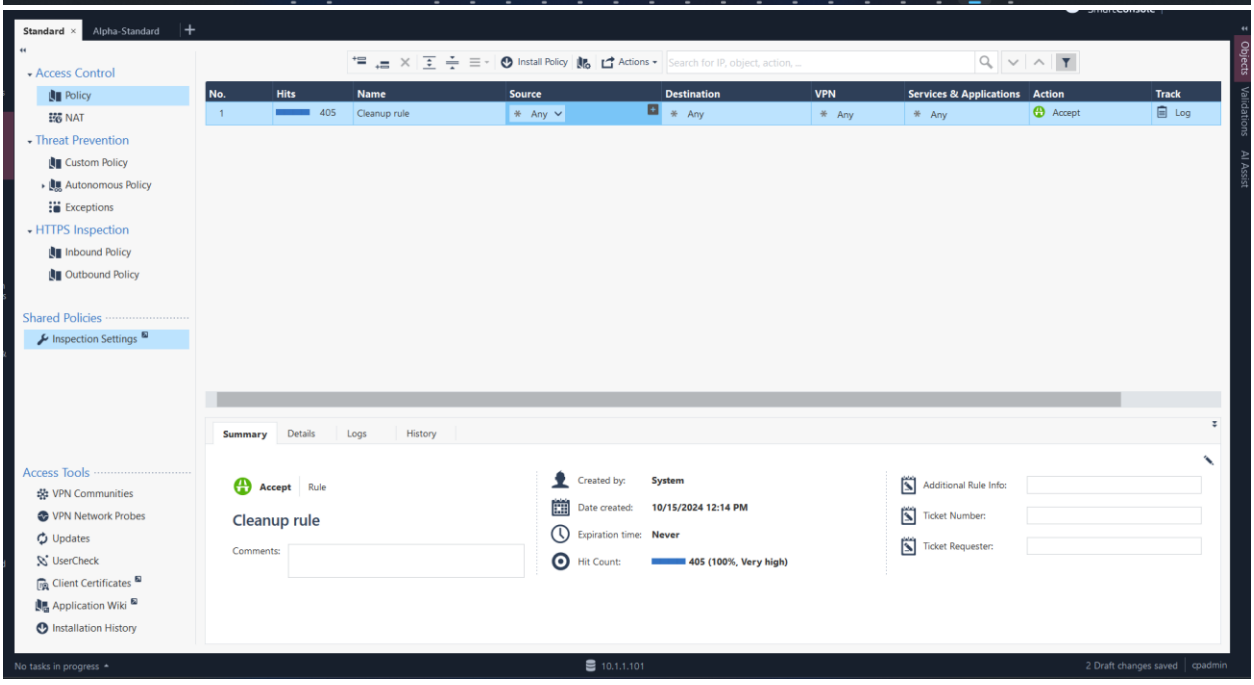
Name: **Ihemdi Pius Emeka**

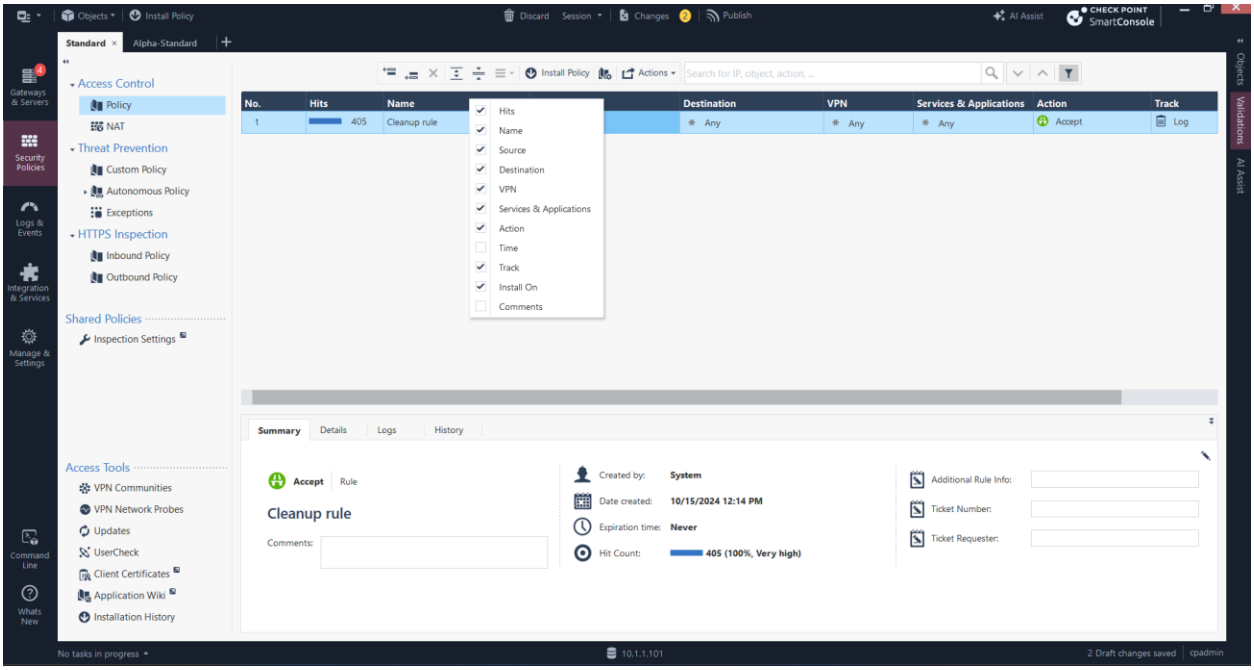
Reg: **C11/26/CCSA/17021**

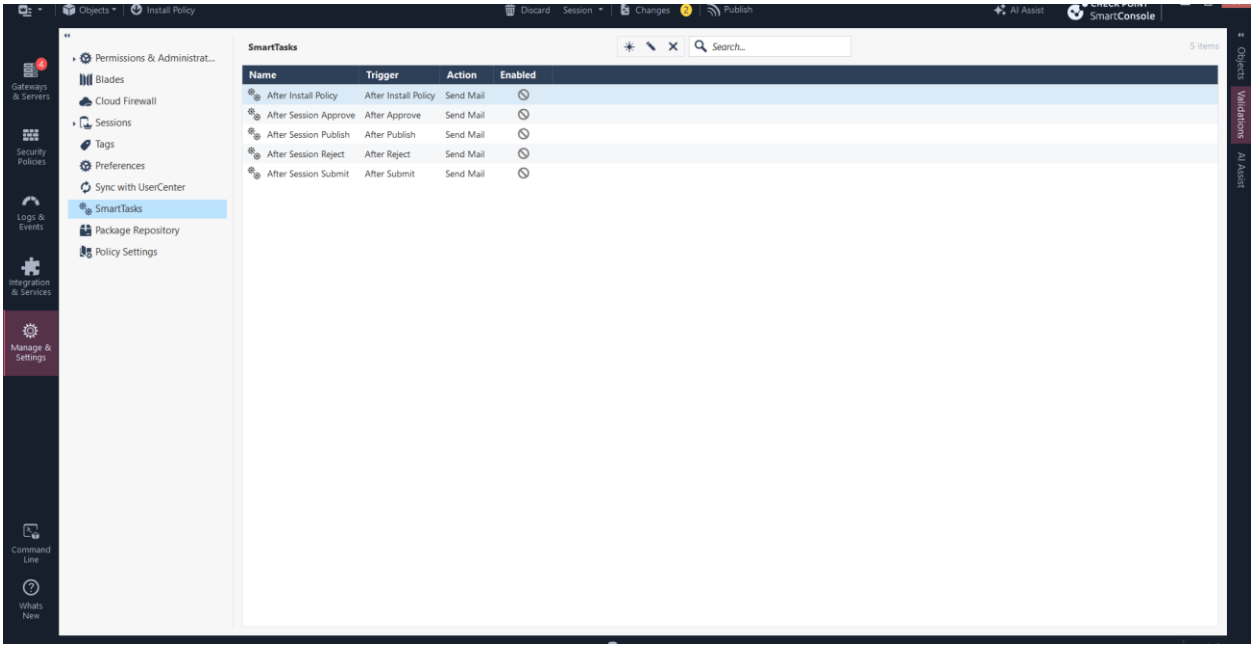
Course: **CCSA104 — Security Policy Management**

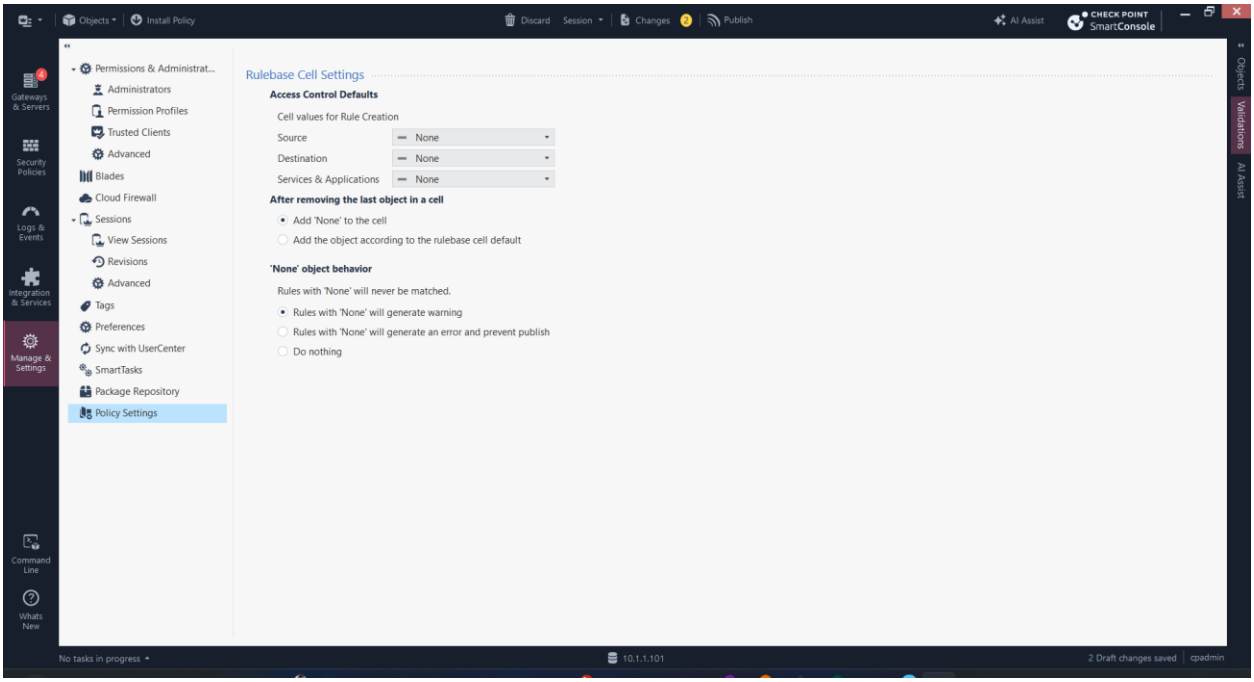
Answers

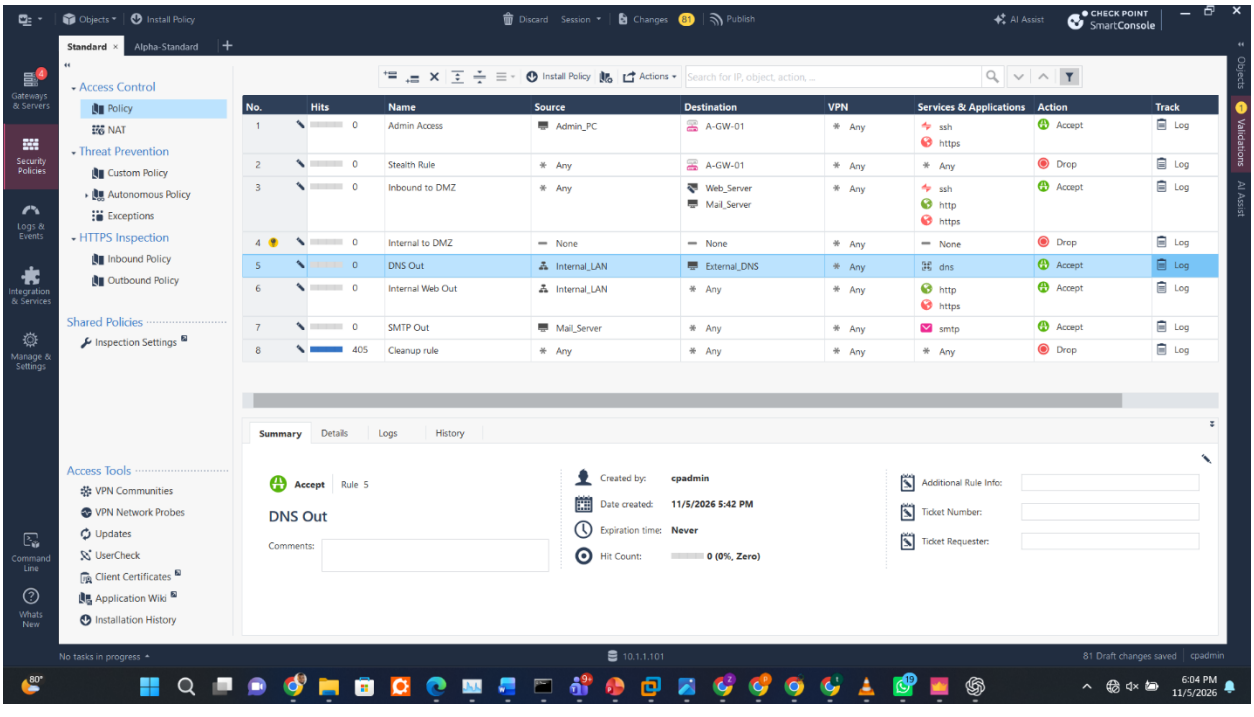


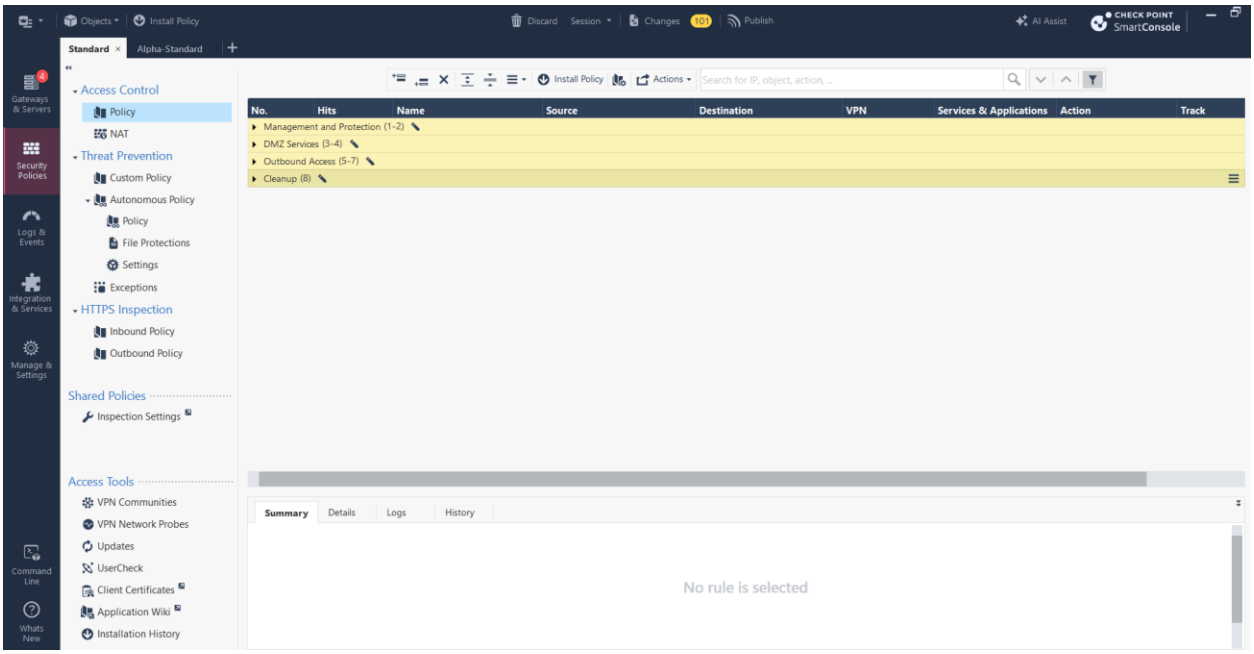


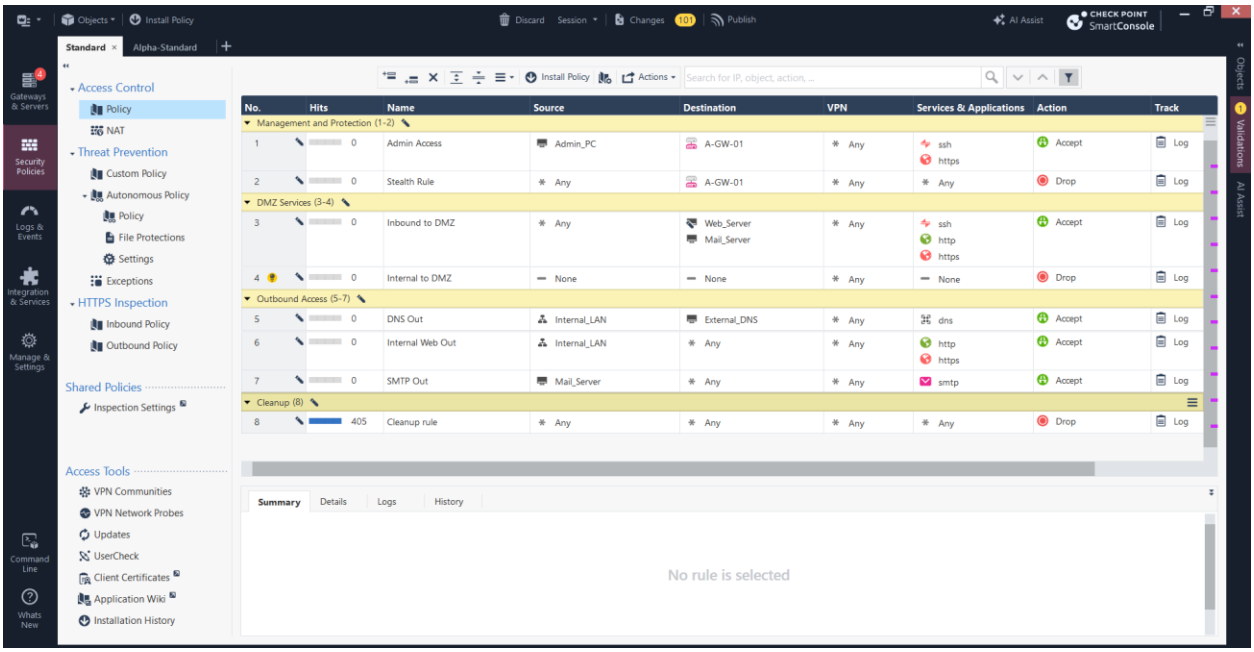

Objects
Install Policy
Discard
Session
Changes
Publish
AI Assist
CHECK POINT SmartConsole
Standard
Alpha-Standard
Gateways & Servers
Security Policies
Logs & Events
Integration & Services
Manage & Settings
Command Line
What's New
Access Control
Policy
NAT
Threat Prevention
Custom Policy
Autonomous Policy
Exceptions
HTTPS Inspection
Inbound Policy
Outbound Policy
Shared Policies
Inspection Settings
Access Tools
VPN Communities
VPN Network Probes
Updates
UserCheck
Client Certificates
Application Wiki
Installation History
Actions
Search for IP, object, action, ...
No.
Hits
Name
Source
Destination
VPN
Services & Applications
Action
Track
Time
Install On
Comments
1
405
Cleanup rule
Any
Accept
Log
Summary
Details
Logs
History
Accept
Rule
Cleanup rule
Comments:
Created by:
System
Date created:
10/15/2024 12:14 PM
Expiration time:
Never
Hit Count:
405 (100%, Very high)
Additional Rule Info:
Ticket Number:
Ticket Requester:
No tasks in progress
10.1.1.101
2 Draft changes saved
cpadmin
Objects
Validations
AI Assist


Objects
Install Policy
Discard
Session
Changes
Publish
AI Assist
CHECK POINT SmartConsole
Gateways & Servers
Security Policies
Logs & Events
Integration & Services
Manage & Settings
Command Line
What's New
Permissions & Administrat...
Blades
Cloud Firewall
Sessions
Tags
Preferences
Sync with UserCenter
SmartTasks
Package Repository
Policy Settings
SmartTasks
Search...
5 items
Name
Trigger
Action
Enabled
After Install Policy
After Install Policy
Send Mail
After Session Approve
After Approve
Send Mail
After Session Publish
After Publish
Send Mail
After Session Reject
After Reject
Send Mail
After Session Submit
After Submit
Send Mail
Objects
Validations
AI Assist

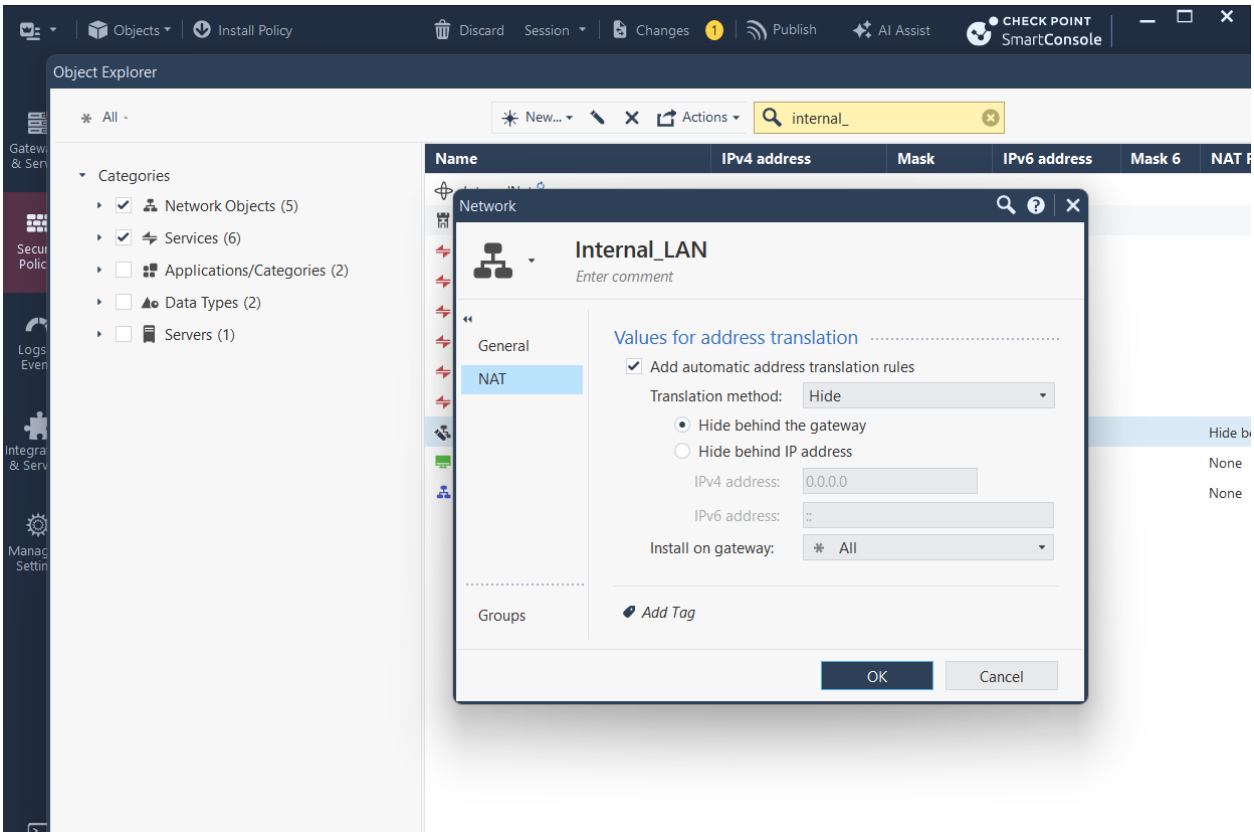

Objects
Install Policy
Discard
Session
Changes
Publish
AI Assist
CHECK POINT SmartConsole
Permissions & Administrat...
Administrators
Permission Profiles
Trusted Clients
Advanced
Blades
Cloud Firewall
Sessions
View Sessions
Revisions
Advanced
Tags
Preferences
Sync with UserCenter
SmartTasks
Package Repository
Policy Settings
Rulebase Cell Settings
Access Control Defaults
Cell values for Rule Creation
Source None
Destination None
Services & Applications None
After removing the last object in a cell
Add 'None' to the cell
Add the object according to the rulebase cell default
'None' object behavior
Rules with 'None' will never be matched.
Rules with 'None' will generate warning
Rules with 'None' will generate an error and prevent publish
Do nothing
Gateways & Servers
Security Policies
Logs & Events
Integration & Services
Manage & Settings
Command Line
What's New
No tasks in progress
10.1.1.101
2 Draft changes saved
cpadmin

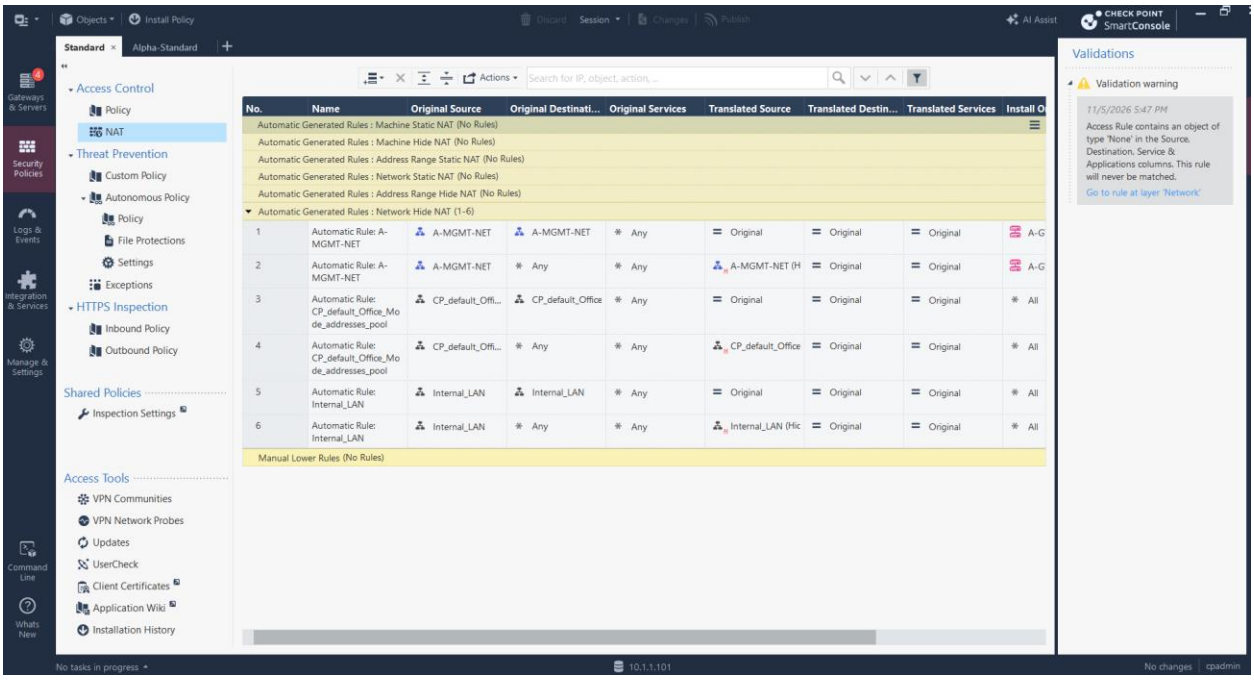

Objects
Install Policy
Discard
Session
Changes
Publish
AI Assist
CHECK POINT SmartConsole
Standard
Alpha-Standard
Access Control
Policy
NAT
Threat Prevention
Custom Policy
Autonomous Policy
Exceptions
HTTPS Inspection
Inbound Policy
Outbound Policy
Shared Policies
Inspection Settings
Access Tools
VPN Communities
VPN Network Probes
Updates
UserCheck
Client Certificates
Application Wiki
Installation History
Actions
Search for IP, object, action, ...
No. Hits Name Source Destination VPN Services & Applications Action Track
1 0 Admin Access Admin_PC A-GW-01 Any ssh https Accept Log
2 0 Stealth Rule Any A-GW-01 Any Any Drop Log
3 0 Inbound to DMZ Any Web_Server Mail_Server Any ssh http https Accept Log
4 0 Internal to DMZ None None Any None Drop Log
5 0 DNS Out Internal_LAN External_DNS Any dns Accept Log
6 0 Internal Web Out Internal_LAN Any Any http https Accept Log
7 0 SMTP Out Mail_Server Any Any smtp Accept Log
8 405 Cleanup rule Any Any Any Any Drop Log
Summary
Details
Logs
History
Accept
Rule 5
DNS Out
Comments:
Created by: cpadmin
Date created: 11/5/2026 5:42 PM
Expiration time: Never
Hit Count: 0 (0%, Zero)
Additional Rule Info:
Ticket Number:
Ticket Requester:
Gateways & Servers
Security Policies
Logs & Events
Integration & Services
Manage & Settings
Command Line
What's New
No tasks in progress
10.1.1.101
81 Draft changes saved
cpadmin
6:04 PM
11/5/2026


Objects
Install Policy
Discard
Session
Changes
101
Publish
AI Assist
CHECK POINT SmartConsole
Standard
Alpha-Standard
Gateways & Servers
Security Policies
Logs & Events
Integration & Services
Manage & Settings
Command Line
Whats New
Access Control
Policy
NAT
Threat Prevention
Custom Policy
Autonomous Policy
Policy
File Protections
Settings
Exceptions
HTTPS Inspection
Inbound Policy
Outbound Policy
Shared Policies
Inspection Settings
Access Tools
VPN Communities
VPN Network Probes
Updates
UserCheck
Client Certificates
Application Wiki
Installation History
Actions
Search for IP, object, action, ...
No.
Hits
Name
Source
Destination
VPN
Services & Applications
Action
Track
Management and Protection (1-2)
DMZ Services (3-4)
Outbound Access (5-7)
Cleanup (8)
Summary
Details
Logs
History
No rule is selected


Objects
Install Policy
Discard
Session
Changes
101
Publish
AI Assist
CHECK POINT SmartConsole
Standard
Alpha-Standard
Gateways & Servers
Security Policies
Logs & Events
Integration & Services
Manage & Settings
Command Line
Whats New
Access Control
Policy
NAT
Threat Prevention
Custom Policy
Autonomous Policy
Policy
File Protections
Settings
Exceptions
HTTPS Inspection
Inbound Policy
Outbound Policy
Shared Policies
Inspection Settings
Access Tools
VPN Communities
VPN Network Probes
Updates
UserCheck
Client Certificates
Application Wiki
Installation History
Actions
Search for IP, object, action, ...
No.
Hits
Name
Source
Destination
VPN
Services & Applications
Action
Track
Management and Protection (1-2)
1 Admin Access Admin_PC A-GW-01 Any ssh https Accept Log
2 Stealth Rule Any A-GW-01 Any Any Drop Log
DMZ Services (3-4)
3 Inbound to DMZ Any Web_Server Mail_Server Any ssh http https Accept Log
4 Internal to DMZ None None Any None Drop Log
Outbound Access (5-7)
5 DNS Out Internal_LAN External_DNS Any dns Accept Log
6 Internal Web Out Internal_LAN Any Any http https Accept Log
7 SMTP Out Mail_Server Any Any smtp Accept Log
Cleanup (8)
8 405 Cleanup rule Any Any Any Any Drop Log
Summary
Details
Logs
History
No rule is selected

Task 6





Theory

Why is this an automatic NAT rule rather than a manual one?

It is an automatic NAT rule because the translation was configured directly in the network object's NAT settings. SmartConsole generates the corresponding NAT rule automatically instead of requiring the administrator to create the translation manually in the NAT rulebase.

What does many-to-one translation mean?

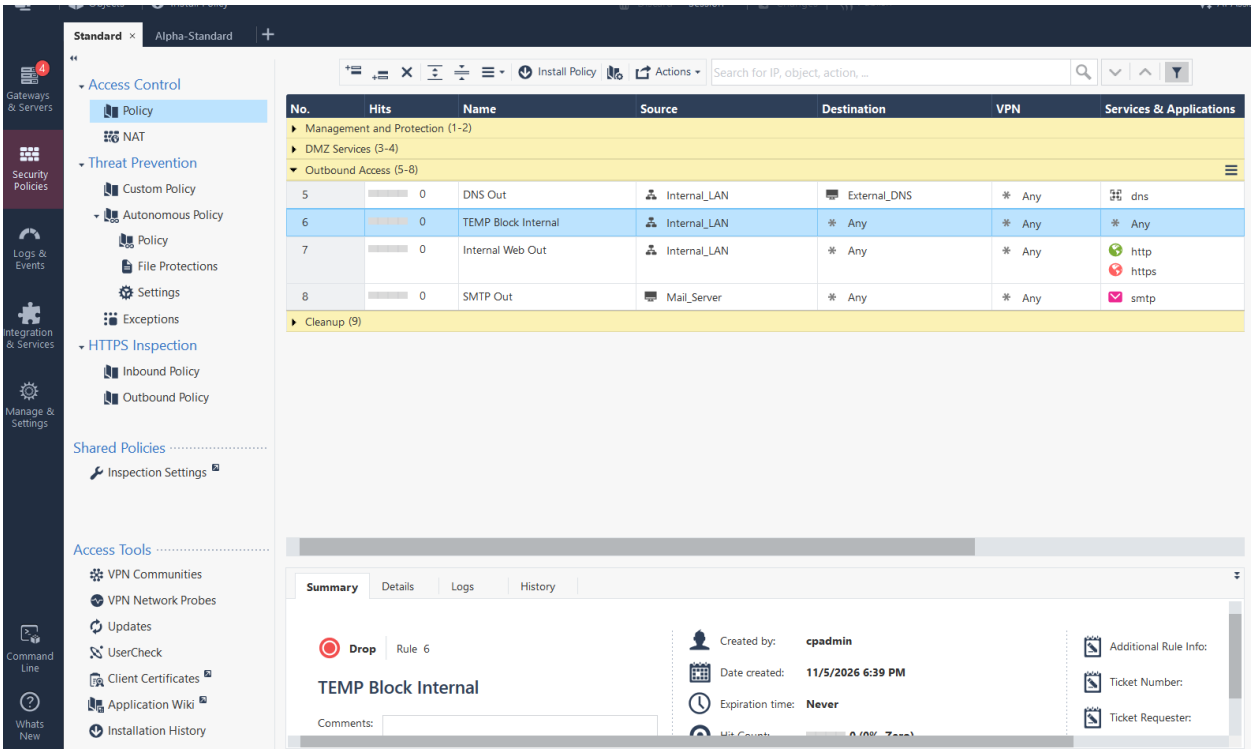
Many-to-one translation means multiple internal devices can share one public or translated IP address when accessing external networks.

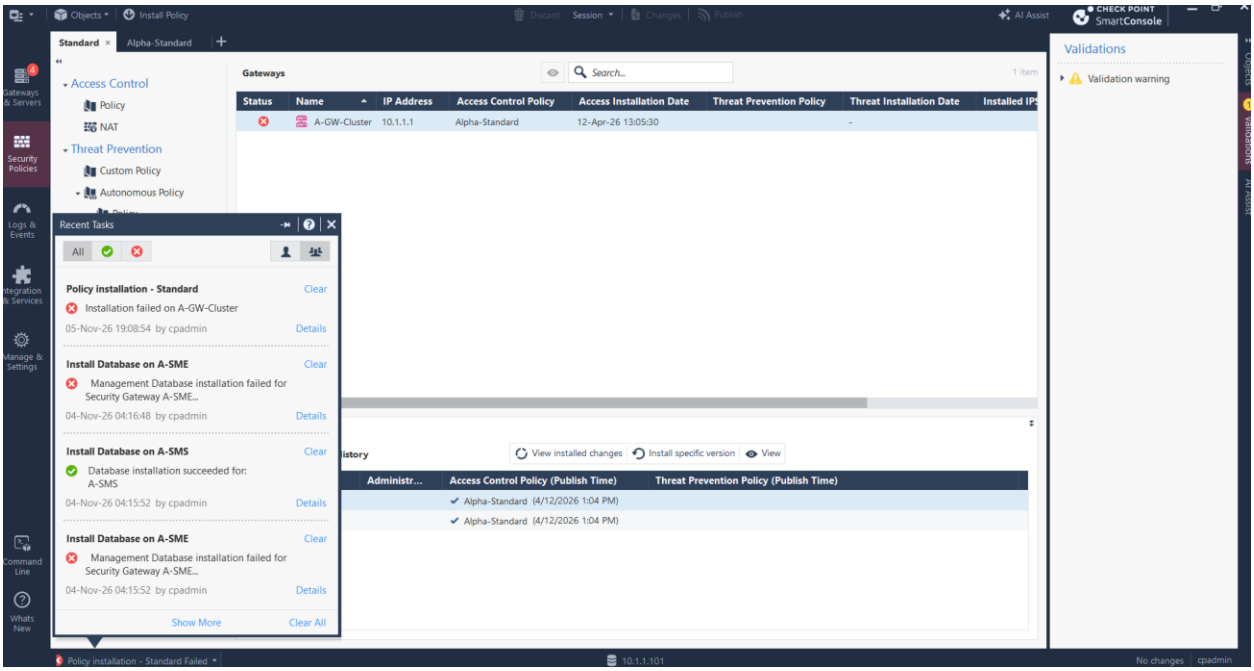
What happens to the source port?

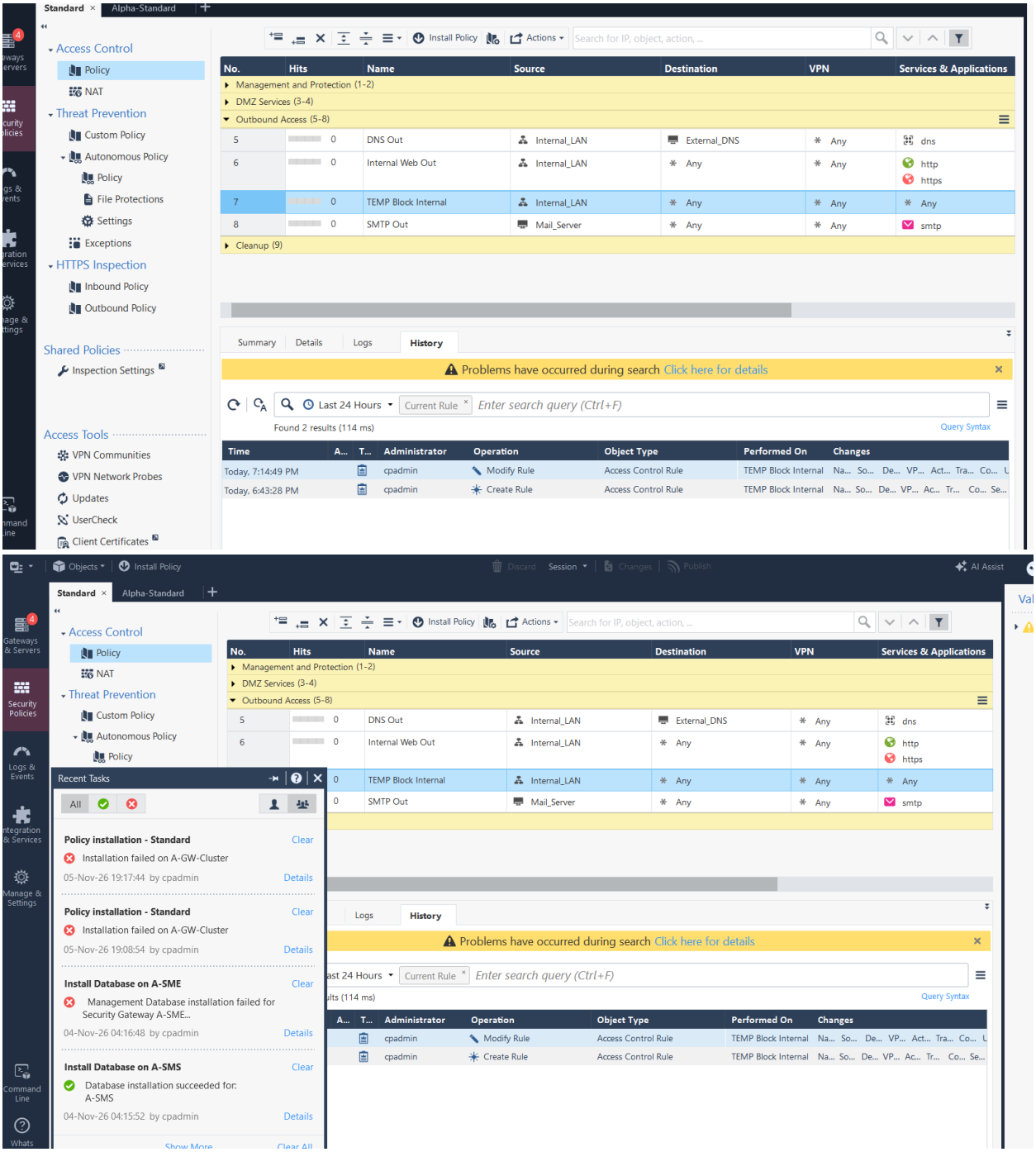
The gateway uses Port Address Translation (PAT) to distinguish connections from different internal devices. It can translate source ports so that multiple connections can share the same translated IP address while maintaining separate connection mappings.

Step 7 – Written explanation

Specific firewall rules should be placed above general rules because Check Point evaluates traffic against rules in order. When TEMP Block Internal was above Internal Web Out, it matched the traffic first and dropped the connection. When it was moved below Internal Web Out, the web access rule matched first and accepted the connection.


Standard
Alpha-Standard
Access Control
Policy
NAT
Threat Prevention
Custom Policy
Autonomous Policy
Policy
File Protections
Settings
Exceptions
HTTPS Inspection
Inbound Policy
Outbound Policy
Shared Policies
Inspection Settings
Access Tools
VPN Communities
VPN Network Probes
Updates
UserCheck
Client Certificates
Application Wiki
Installation History
Install Policy
Actions
Search for IP, object, action, ...
No.
Hits
Name
Source
Destination
VPN
Services & Applications
Management and Protection (1-2)
DMZ Services (3-4)
Outbound Access (5-8)
5 0 DNS Out Internal_LAN External_DNS Any dns
6 0 TEMP Block Internal Internal_LAN Any Any Any
7 0 Internal Web Out Internal_LAN Any Any http https
8 0 SMTP Out Mail_Server Any Any smtp
Cleanup (9)
Summary
Details
Logs
History
Drop
Rule 6
TEMP Block Internal
Comments:
Created by: cpadmin
Date created: 11/5/2026 6:39 PM
Expiration time: Never
Additional Rule Info:
Ticket Number:
Ticket Requester:


Gateways
Search...
Status
Name
IP Address
Access Control Policy
Access Installation Date
Threat Prevention Policy
Threat Installation Date
A-GW-Cluster 10.1.1.1 Alpha-Standard 12-Apr-26 13:05:30
Validations
Validation warning
Recent Tasks
Policy installation - Standard
Installation failed on A-GW-Cluster
05-Nov-26 19:08:54 by cpadmin
Install Database on A-SME
Management Database installation failed for Security Gateway A-SME...
04-Nov-26 04:16:48 by cpadmin
Install Database on A-SMS
Database installation succeeded for: A-SMS
04-Nov-26 04:15:52 by cpadmin
Install Database on A-SME
Management Database installation failed for Security Gateway A-SME...
04-Nov-26 04:15:52 by cpadmin
Show More
Clear All
View installed changes
Install specific version
View
Access Control Policy (Publish Time)
Threat Prevention Policy (Publish Time)
Alpha-Standard (4/12/2026 1:04 PM)
Alpha-Standard (4/12/2026 1:04 PM)


Standard
Alpha-Standard
Access Control
Policy
NAT
Threat Prevention
Custom Policy
Autonomous Policy
Policy
File Protections
Settings
Exceptions
HTTPS Inspection
Inbound Policy
Outbound Policy
Shared Policies
Inspection Settings
Access Tools
VPN Communities
VPN Network Probes
Updates
UserCheck
Client Certificates
Install Policy
Actions
Search for IP, object, action, ...
No. Hits Name Source Destination VPN Services & Applications
Management and Protection (1-2)
DMZ Services (3-4)
Outbound Access (5-8)
5 0 DNS Out Internal_LAN External_DNS Any dns
6 0 Internal Web Out Internal_LAN Any Any http https
7 0 TEMP Block Internal Internal_LAN Any Any Any
8 0 SMTP Out Mail_Server Any Any smtp
Cleanup (9)
Summary Details Logs History
Problems have occurred during search Click here for details
Last 24 Hours
Current Rule
Enter search query (Ctrl+F)
Found 2 results (114 ms)
Query Syntax
Time A... T... Administrator Operation Object Type Performed On Changes
Today, 7:14:49 PM cpadmin Modify Rule Access Control Rule TEMP Block Internal
Today, 6:43:28 PM cpadmin Create Rule Access Control Rule TEMP Block Internal

Objects
Install Policy
Discard
Session
Changes
Publish
AI Assist
Recent Tasks
All
Policy installation - Standard
Clear
Installation failed on A-GW-Cluster
05-Nov-26 19:17:44 by cpadmin
Details
Policy installation - Standard
Installation failed on A-GW-Cluster
05-Nov-26 19:08:54 by cpadmin
Install Database on A-SME
Management Database installation failed for Security Gateway A-SME...
04-Nov-26 04:16:48 by cpadmin
Install Database on A-SMS
Database installation succeeded for: A-SMS
04-Nov-26 04:15:52 by cpadmin
Show More
Clear All

Multi-choice Question

Correct option

Q1 B

Q2 A

Q3 C

Q4 D

Q5 B

Section B2: Short Answer Questions (15 marks)

Q6. Distinguish between Explicit rules and Implied rules. State where Implied rules are configured and where they can be placed. What is the Implicit Cleanup Action?

Answer:

Explicit rules are rules created manually by the administrator in the Access Control rulebase. They specify which traffic should be allowed or blocked.

Implied rules are automatically generated by Check Point to support essential firewall operations, such as certain management and control traffic. They are configured through Global Properties in SmartConsole and can be positioned before or after the explicit rules, depending on their configuration.

The Implicit Cleanup Action is the default action applied to traffic that does not match an applicable rule. It generally drops the traffic, although its behaviour depends on the configured global properties.

Q7. Explain why rule order matters in a rulebase. Give one example from your own rulebase.

Answer:

Rule order matters because the firewall evaluates traffic against the rules from top to bottom. When traffic matches a rule, that rule determines whether the connection is accepted or dropped.

For example, if TEMP Block Internal is placed above Internal Web Out, internal web traffic will be dropped because it matches the temporary blocking rule first. If the blocking rule is moved below Internal Web Out, the web traffic will be accepted by the earlier rule.

Q8. Explain the difference between saving and publishing changes in SmartConsole. What happens to unpublished changes when you install a policy, and why do many organisations publish only during a change window?

Answer:

Saving changes keeps my configuration work within my SmartConsole session, while publishing commits the changes to the Security Management Server so they become part of the management database.

Unpublished changes are not included when a policy is installed. The gateway receives the last published policy selected for installation.

Many organisations publish during an approved change window to control when configuration changes are committed, reduce the risk of unexpected changes and allow administrators to monitor the network after implementation.

Q9. Compare Global Properties with Policy Packages. Give one example of what each controls and say how you open each in SmartConsole.

Answer:

Global Properties control general security management settings that can affect firewall behaviour across the environment. For example, they can control implied firewall rules and other global firewall settings. They can be opened through the Global Properties option under the management or settings area of SmartConsole.

Policy Packages contain the security rules and policy layers that are assigned to specific gateways. For example, a policy package can define which internal users are allowed to access the Internet. They can be opened through Security Policies or Manage Policies and Layers in SmartConsole.

Q10. Name the four predefined Security Zones. State two rules about assigning gateway interfaces to zones, and one benefit of using zone-based rules.

Answer:

The four predefined Security Zones are:

1. Internal
2. External
3. DMZ
4. VPN

Two rules for assigning gateway interfaces to zones are:

- Each gateway interface should be assigned to the appropriate security zone according to the network it connects to.
- An interface should not be assigned conflicting zone identities; its zone assignment must correctly represent its network connection.

One benefit of zone-based rules is that they make security policies easier to manage because administrators can control traffic between network segments using their security zones rather than creating separate rules for every individual network.

Note: Confirm the precise interface-assignment restrictions and zone names in your SmartConsole version and Lab 4A guide, as these details are not specified in the assignment itself.

Section B3: Scenario Questions (10 marks)

Each question carries 5 marks.

Q11. A new administrator inserts a rule with Source Any, Destination Any, Services Any and Action Accept above the Stealth rule. Describe two security problems this causes, explain how to correct it, and name the type of rule that is allowed to sit above the Stealth rule and why.

Answer:

The first security problem is that the rule allows traffic from any source to any destination, making the firewall too permissive. This can allow unauthorised traffic to reach the gateway.

The second problem is that it defeats the purpose of the Stealth rule, which is designed to protect the gateway from unauthorised access. Attackers may be able to reach services running on the gateway.

To correct the problem, the administrator should delete the unrestricted Accept rule and place the Stealth rule immediately after the authorised management rules.

Only specific management rules should be placed above the Stealth rule. For example, the Admin Access rule allows authorised administrators to manage the gateway through SSH and HTTPS. This ensures that legitimate management access is maintained while other traffic to the gateway is blocked.

Q12. After Hide NAT is configured for Internal_LAN, staff report they cannot reach the Internet.

(a) State two benefits of NAT. (2 marks)

Answer:

1. NAT allows multiple internal devices to share a single public IP address, helping to conserve public IP addresses.
2. NAT hides the internal network addressing scheme from external networks, making the internal network structure less directly exposed.

(b) Describe three checks you would make to find the fault, and explain why each one matters. (3 marks)

Answer:

1. Check the NAT configuration: I would confirm that automatic Hide NAT is enabled for Internal_LAN and that the network is hidden behind the correct gateway address. This ensures that internal traffic is translated before leaving the network.

2. Check the firewall policy: I would verify that the DNS Out and Internal Web Out rules allow the required traffic and that the rules are installed on the gateway. This ensures that DNS and web connections are not being blocked by the firewall.

3. Check gateway connectivity and routing: I would verify that the gateway has a working external interface, a valid default route and connectivity to the Internet. This helps determine whether the problem is caused by routing or an upstream network connection rather than the NAT configuration.